

UMBILICAL CORD PROLAPSE

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Session objectives:

- ✓ At the end of this session you are expected to:
 - Define umbilical cord prolapse
 - Describe type of umbilical cord prolapse
 - Identify the diagnosis of umbilical cord prolapse
 - Discuss the management of umbilical cord prolapse

➤ Definition

- umbilical cord prolapse is when umbilical cord (UC) descends alongside or beyond the fetal presenting part in the presence of ruptured membranes

➤ Classification

- ❖ **Overt UC prolapse:** Protrusion of the UC in advance of the fetal presenting part with ruptured fetal membranes
- ❖ **Occult UC prolapse:** Cord descends alongside, but not past, the presenting part with intact / ruptured fetal membranes
- ❖ **Cord presentation:** Prolapse of UC below the level of the presenting part with intact fetal membranes

➤ Risk factors

General

- Malpresentations
- Unengaged presenting part
- Prematurity
- Multifetal gestation
- PROM
- Abnormal placentation
- Multiparity
- Polyhydramnios
- Long UC
- Congenital anomalies
- Low birth weight less than 2.5 kg.

Procedure related

- ARM with unengaged fetal presenting part
- Vaginal manipulation of the fetus with ruptured membranes
- ECV

➤ Diagnosis

❖ Overt UCP:

- Presence of palpable cord (pulsatile or non-pulsatile) on pelvic examination or visible cord outside the introitus

❖ Occult UCP

- Presence of severe prolonged fetal bradycardia or
- moderate to severe variable decelerations after a previous normal tracing on CTG/ Pinnard stethoscope or fetal death

❖ Cord presentation: Loops of cord are palpated through the fetal membranes on digital vaginal examination or

❖ seen in front of the presenting part on ultrasound examination

➤ Management

❖ General measures

- Call for assistance
- Secure IV fluid
- Check for cord pulsation. If absent, confirm fetal heart beat with fetoscope/ U/S
- Prepare for resuscitation of the newborn
- In cord presentation, do not rupture fetal membranes at any stage of labor; deliver the fetus by CS
- Monitor FHB while preparing for delivery

Cont...

- ✓ If the woman is in the first stage of labor;
 - Push the presenting part up
 - Place the other hand on the abdomen in the suprapubic region to keep the presenting part out of the pelvis
 - Keep the hand on the abdomen until a caesarean section can be performed
 - If available, give tocolytics

Cont...

- Do not manipulate the cord
- Avoid exposure of the cord to cold environment to avoid cord spasm
- Bladder filling
- Perform immediate caesarean delivery
- Client should be placed in Trendelenberg or knee-chest position

Cont...

- ✓ If the woman is in the second stage of labor;
 - Expedite vaginal birth if deemed quicker than cesarean section
 - Obstetric vacuum is preferable over forceps, if prerequisites are met
 - If there is malpresentation or if prerequisites for instrumental delivery are not fulfilled, deliver immediately by CS
- ✓ *NOTE: Delivery should be accomplished within 30 minutes from the time of diagnosis*

CEPHALOPELVIC DISPROPORTION(CPD)

➤ Definition

- CPD is disparity between the fetal head and maternal pelvis which leads to inability of the fetal head to pass through the maternal pelvis for mechanical reasons

➤ Causes of CPD (clinical classification)

❖ **Absolute CPD:** true mechanical obstruction as a result of

- **Permanent (maternal) factors:** contracted pelvis (commonest), pelvic exostoses, spondylolisthesis, anterior sacro-coccygeal tumors; or

- **Temporary (fetal) factors:** Hydrocephalus, large fetus

Cont...

❖ **Relative CPD:** where the fetus may be delivered vaginally if a favorable combination of other factors can be achieved

E.g. face presentation and occipito-posterior positions
(rotation / flexion of the head may occur during labor progress)

➤ **Diagnosis of CPD**

History

- Prolonged labor
- Previous history of prolonged labor
- Previous history of perinatal death
- Previous history of obstetric trauma
- Properly documented obstetric record

physical examination

- Abnormal progress of labor
- Abnormal clinical pelvimetry
- Molding
- Caput Succedaneum
- Abnormal degree of head flexion
- Asynclitism
- Fetal Macrosomia
- Fetal distress

➤ **Management of CPD**

❖ **Trial of labor:**

- ✓ Trial of labor is conducted in a woman with suspected CPD
- ✓ CPD is suspected if there is;
 - previous history of prolonged labor with bad obstetric history
 - if the parturient is teenage
 - if the cervicogram is crossing the alert line or presence of borderline pelvis
 - Borderline pelvis is entertained if the obstetric conjugate is 8 to 10 cm
- ✓ It is done in an equipped and staffed hospital for operative procedures
- ✓ It is the best diagnostic approach if there is no other risk factor (such as previous CS),
- ✓ The trial continues as long as labor progresses , fetal and maternal status are well

Cont...

❖ Caesarean delivery;

✓ Caesarean delivery indicated;

- In permanent absolute disparities (e.g. severe pelvic contracture (OC of 6-8 cm) or extreme pelvic contracture (OC < 6 cm))
- Macrosomic fetus greater than 4.5 kg (4.0 kg if the mother is diabetic)
- Fetal hydrocephalus may be managed by cephalocentesis
- Craniotomy is indicated if the fetus is dead and prerequisites for destructive delivery are fulfilled

➤ **Complications of CPD**

Maternal

- Prolonged labor
- obstructed labor
- PPH
- Maternal sepsis

Fetal / neonatal

- Fetal distress
- Perinatal asphyxia
- Neonatal infections
- Perinatal death

Obstructed labor

- **Learning objectives**
- ✓ Define obstructed labor
- ✓ list the causes of obstructed labor
- ✓ Describe the clinical features of obstructed labor
- ✓ Explain the management of obstructed labor
- ✓ List the complications of obstructed labor

Cont...

➤ Definition

- obstructed labor is failure of progressive descent of the presenting part of the fetus in the birth canal for mechanical reasons in spite of good uterine contraction.
- It is due to mechanical factors not to disturbance of uterine physiology
- It is an absolute condition, further progress is impossible

➤ Causes of OL

Maternal (Passage)

- Contracted pelvis / CPD
(commonest)
- Soft tissue abnormalities
(e.g. tumor previa, vaginal
septum, tight perineum,
uterine congenital anomalies)

Fetal (Passenger)

- Malpresentations
- Macrosomia
- Malposition
- Locked twins,
- conjoined twins
- Fetal anomalies e.g.
hydrocephalus
- Shoulder dystocia

➤ **Incidence and importance of OL**

- Most common causes of maternal and perinatal mortality and morbidity in developing country
- Globally, OL is responsible for 8% of all maternal deaths
- It accounts for 22% of the total maternal death in Ethiopia
- It is mainly related to the availability, accessibility and quality of antepartum and intrapartum services
- OL is considered as a sign of major failure in obstetric care

➤ **Clinical feature**

❖ **History**

- Prolonged labor
- Prolonged rupture of membrane
- medical history(rickets, osteomalacia, or pelvic injury)

❖ **On physical examination**

- Maternal and fetal distress
- Ketoacidosis
- Fever
- Abdominal pain.
- Dehydration
- Shock
- High head
- Frequent, long, strong uterine contraction
- sign of uterine rupture
- Bandl's ring(sign of three tumor abdomen)

➤ **Vaginal examination**

- Foul smelling vaginal discharge
- Hot, dry vagina
- Amniotic fluid drained away
- concentrated urine which contain blood or meconium
- edematous cervix
- cervix poorly applied to the presenting part
- Large caput succedaneum
- Excessive molding
- Prolapsed arm, face, brow...

Cont..

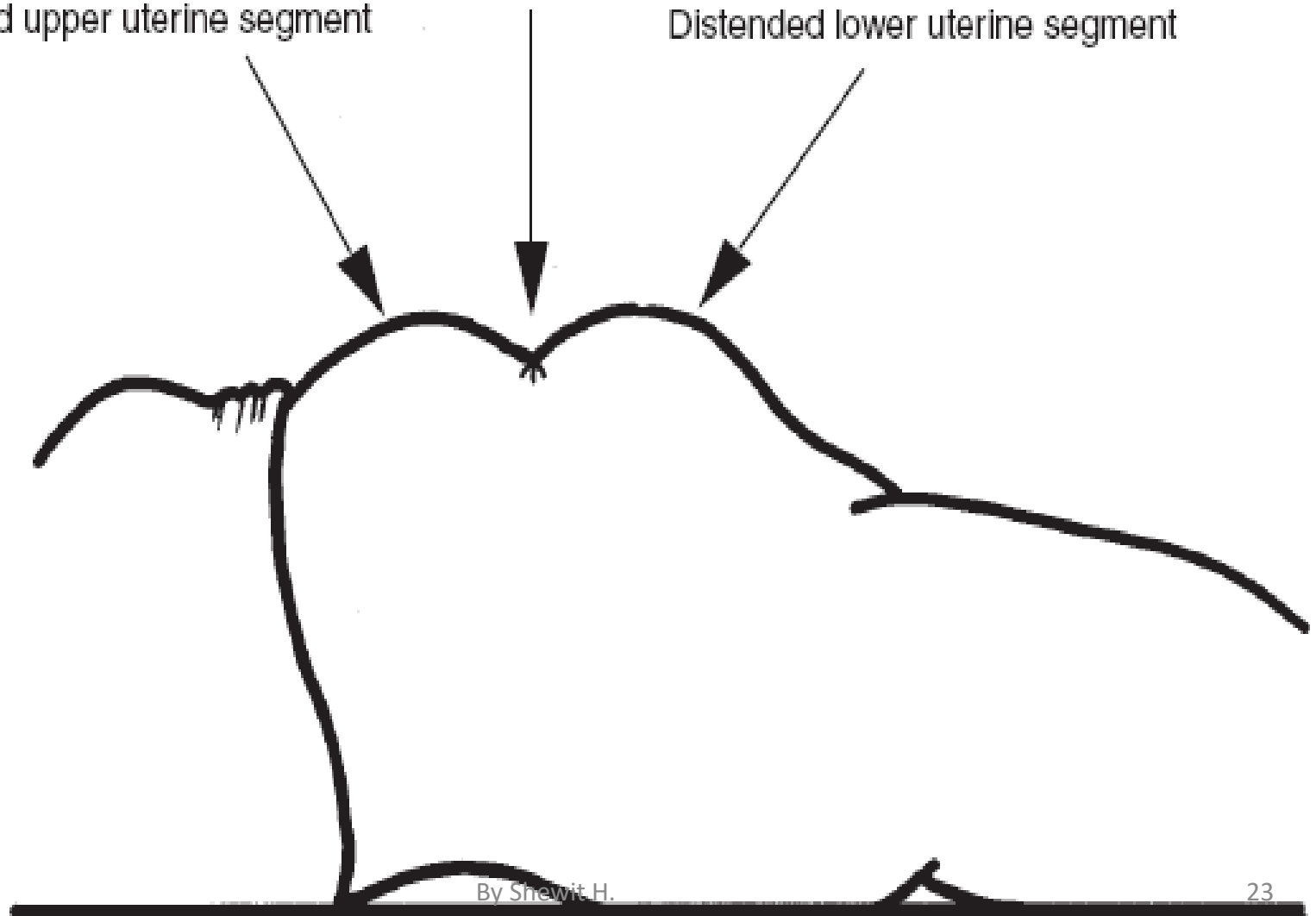
- ✓ The three tumor on the abdomen are due to;
 - Grossly thickened and retracted upper uterine segment above Bandl's ring
 - Thinly distended lower uterine segment below the ring and
 - Fully distended /edematous bladder distending the lower abdomen with the Bandl's ring in between
- ✓ (Bandl's ring is a late sign of obstructed labor
- ✓ It can be seen as a depression across the abdomen at about the level of the umbilicus.
- ✓ It is a warning sign of an impending uterine rupture.

Shape of the abdomen in obstructed labour

Bandl's ring

Retracted upper uterine segment

Distended lower uterine segment



➤ **Management**

✓ Initial management of OL involves;

- Resuscitation
- monitoring of the life endangering conditions such as shock and sepsis followed by identification and treatment of the cause of OL and its complications.
- Send an urgent blood sample
- Give broad spectrum antibiotics
- Give analgesics

❖ **Delivery**

✓ Alive fetus;

- Perform emergency cesarean delivery

✓ Dead fetus with imminent signs of uterine rupture;

- perform cesarean delivery

✓ Dead fetus without imminent signs of uterine rupture

- perform destructive delivery if prerequisites are fulfilled

➤ **Complications**

Early complications

- Atonic PPH
- uterine rupture
- Peripartum infection
- maternal death, fetal distress
- fetal & neonatal infections
- fetal and neonatal death.

Late complications

- Fistula (e.g. vesico-vaginal, rectovaginal) and its aftermath
- infertility
- psychological trauma due to the painful labor experience

➤ **Prevention of OL**

- Good nutritional supply since childhood
- Avoid early marriage
- Universal ANC

UTERINE RUPTURE

➤ Definitions

- ✓ **Uterine rupture** is tear through the uterine wall above the cervico-uterine junction during pregnancy and labor
- ✓ **Silent uterine rupture** is rupture of the uterus before the onset of labor
 - It usually occurs in patients with previous uterine scar involving the upper uterine segment

➤ Risk factors

- Obstructed labor and
- previous C/S scar are the most common risk factors

➤ **Classification of uterine rupture**

- ❖ **Complete:** All the **three layers** of the uterus are involved and there is a direct communication between the uterine and abdominal cavities
- ❖ **Incomplete:** the **peritoneum** covering the uterus remains intact

The classic s/s of **complete rupture**

- Sudden onset abdominal pain
- Cessation of uterine contractions
- Easily palpable fetal parts
Abnormal uterine contour
- Signs of intra-abdominal hemorrhage
- Tender abdomen
- Absent fetal heart sounds
- Vaginal bleeding
- Recession of presenting part
- Hemorrhagic shock

Clinical finding of **incomplete rupture**

- Rapid maternal pulse
- Labor pain may continue
- FHR abnormalities is the most reliable warning sign
- Vaginal bleeding

➤ **Management of uterine rupture**

❖ **Emergency management**

- Secure IV lines bilaterally with large bore cannula
- Resuscitation with IV fluids and blood products
- Prepare for operative interventions (e.g. determine hematocrit, blood group and RH, avail cross-matched blood and organize the OR)

❖ **Surgical intervention**

- Total hysterectomy
- Subtotal hysterectomy
- Repair of uterine tear with preservation of fertility

Preterm labour

Objectives:

- Definition of preterm labor
- Discuss etiology of preterm labor
- Describe diagnosis of preterm labor
- List complications of preterm labor
- Outline the managements of preterm labor

Cont...

- ✓ Preterm labor is onset of labor after 28th weeks and before 37 completed week of gestational age
- ✓ Affects 5- 10% of pregnancies on average
- ✓ responsible for prematurity and
- ✓ 80-90% of perinatal mortality in the developing world

➤ Risk factor

- PROM
- APH
- Systemic infection (anemia, malaria,...)
- Pre Eclampsia / eclampsia
- Uterine abnormality(cervical incompetency,myoma...)
- Trauma
- Uterine Over distention (polyhydramnios, Multiple px
- Placental insufficiency
- RH isommunization
- Malnutrition
- chorioamnionitis
- Emotional stress
- Diabetes mellitus
- Smoking
- Alcoholism and drug abuse
- Cocaine, heroin use

➤ **Diagnosis of Preterm Labor**

- **Confirmed Gestational Age <37 completed weeks** and
- Regular,painful,rhythmic uterine contractions occur at least twice in 10 minut for 30sec
- A cervical dilatation of at least 2 cms or
- Ruptured membranes or
- Cervical effacement of $> 80\%$ or
- Vaginal bleeding

➤ **Management of preterm labor**

✓ Management of preterm labor depends on;

- Gestational age
- cervical dilatation and effacement
- Condition of mother and fetus
- Membrane status(rupture or intact)

✓ Management Options:

- Arresting preterm labor
- Continue preterm labor

Cont...

- ✓ Arresting labor if, Gestational age is less than 34wk
 - Fetus and mother are at good condition
 - Cervical dilatation is less than 4cm
 - cervical effacement is less than 80% and is intact membrane with no bulge
- ✓ Administere tocolytic drugs to arrest & delay labor for as long as possible
- ✓ Could be short term tocolysis for few days(at least one week) until lung maturation agents are administer

Cont...

✓ Continue preterm labor if;

- Sever preeclampsia and eclampsia
- Maternal heamorrhage,cardiac problem
- Cervical dilatation greater than 4cm
- Cervical effacement greater than 80%
- Sever IUGR
- Gestational age near to term
- Fetal death
- Chorioamninitis
- Ruptured or bulging membrane

➤ **Complication of preterm labor**

❖ **Maternal**

- Increased risk of caesarean delivery
- Complications of tocolytic drugs

❖ **Neonatal/Fetal**

- Feeding problems (sucking reflex)
- Neonatal infection and sepsis
- RDS
- Birth trauma
- Hypothermia
- Hypoglycemia
- hypocalcaemia
- hypomagnesaemia,
- Hyperbilirubinemia
- Perinatal asphyxia
- Long term neurologic sequel

PREMATURE RUPTURE OF MEMBRANE (PROM)

➤ **Definitions :** PROM is rupture of membranes before the onset of labor

➤ **Classification**

- **Term PROM:** is rupture of membranes at or after 37 completed weeks of gestation
- **Preterm PROM:** is rupture of membranes before 37 completed weeks of gestation
- **Prolonged PROM :** is rupture of membranes for > 12 hours.

➤ **Risk factors of PROM**

- ❖ **Mechanical factors:** Multifetal gestation, polyhydramnios, cervical conization
- ❖ **Urogenital infections:** UTI, cervicitis, GBS, bacterial vaginosis
- ❖ **Previous history** of preterm PROM, preterm labor
- ❖ **Second and third trimester bleeding** (e.g. abruptio placenta)
- ❖ **Other risk factors:** Low socioeconomic status, nutritional deficiencies, low BMI, smoking and connective tissue disorders

➤ **Diagnosis of PROM**

- ❖ **History:** The classic clinical presentation of PROM is a sudden "gush" of clear or pale-yellow fluid from the vagina
 - Patients can present with intermittent or constant leaking of small amounts of within the vagina
- ❖ **Sterile speculum examination:** Observe amniotic fluid coming out of the cervical canal
- ❖ **Ferning test:** Obtain fluid by swabbing the posterior fornix
 - Spread some fluid on a slide & let it dry for at least 10 minutes
 - Examine it with a microscope and look for a fern-leaf pattern

Cont...

❖ Nitrazine paper test:

- Hold a piece of nitrazine paper in a artery forceps & touch it against the fluid pooled on the speculum blade
- A change from **yellow to blue** indicates presence of amniotic fluid

❖ Pad test

- Place a vaginal pad over the vulva & examine it one hour later visually & by odor
- Wetting with no urine and no vaginal discharge (vaginitis) may suggest PROM

❖ Ultrasound examination: Performed to look for reduction of

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amniotic fluid volume

Approachs to management of PROM

1. Confirm the diagnosis of ROM
2. Evaluate for the presence of chorioamnionitis and labor
3. Determine the gestational age and evaluate the fetal condition
4. Subsequent management based on the above findings

Cont..

- **Indications for expedite delivery:** onset of labor, gestation age ≥ 37 wks, evidence for non-reassuring fetal status, and chorioamnionitis,
- **Expectant management:** Indicated if gestational is below 34 weeks and both the fetal and maternal conditions are stable
 - Advise bed-rest and avoid digital cervical (pelvic) examination
 - Administer corticosteroids for lung maturity
 - do not administere corticosteroid for women with chorioamnionitis

➤ **Complications of PROM**

❖ **Maternal:**

- Abruptio placentae,
- chorioamnionitis, sepsis, higher risk for cesarean delivery
- retained placenta and hemorrhage

❖ **Fetal and Neonatal:**

- Infection,
- umbilical cord compression as a result of oligohydramnios, cord prolapse
- RDS, and pulmonary hypoplasia

References

- Current diagnosis & treatment Gyna & Obes
- Williams Obstetrics 23edn
- Essential Obstetric & Gynecology

THANK YOU